

Future Work and Advanced Model Roadmap

The current phase of this project focused on traditional machine learning algorithms for sentiment classification using Amazon Fine Food Reviews. The following models have been completed and evaluated:

- Logistic Regression
- Multinomial Naive Bayes
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest

These models established a strong baseline for understanding customer sentiment and provided valuable insights into the strengths and limitations of different classification techniques.

Planned Models for Further Development

6. AdaBoost

AdaBoost (Adaptive Boosting) will be implemented to investigate whether sequential boosting can improve sentiment classification performance. Unlike Random Forest, which builds trees independently, AdaBoost focuses on correcting the mistakes made by previous learners. This model is expected to improve classification accuracy by giving more attention to difficult reviews that are frequently misclassified.

7. XGBoost / LightGBM

XGBoost and LightGBM represent state-of-the-art gradient boosting algorithms and are widely used in industry and machine learning competitions. These models are expected to provide stronger predictive performance, improved handling of complex feature interactions, and enhanced robustness when compared with traditional tree-based methods.

Key expected contributions:

- Improved classification accuracy
- Better handling of large-scale datasets
- Strong feature importance analysis
- Faster training and inference (particularly with LightGBM)

8. Long Short-Term Memory (LSTM)

LSTM is a deep learning architecture specifically designed for sequential data. Unlike traditional machine learning algorithms that treat words as independent features, LSTM processes text as sequences and can capture contextual relationships between words.

Expected advantages include:

- Understanding word order and context
- Better handling of linguistic patterns
- Improved representation of customer opinions
- Enhanced sentiment prediction for complex reviews

9. BERT / DistilBERT

BERT (Bidirectional Encoder Representations from Transformers) represents one of the most influential advancements in Natural Language Processing. Unlike traditional machine learning models, BERT uses pre-trained language representations and understands contextual meaning from both directions of a sentence.

Expected contributions:

- State-of-the-art sentiment classification performance
- Superior understanding of context and language nuances
- Better handling of ambiguous and complex customer reviews
- Stronger generalization capabilities across unseen data

DistilBERT may also be explored as a lighter and faster alternative while retaining much of BERT's predictive power.

Expected Project Evolution

The progression from traditional machine learning models to advanced deep learning and transformer-based architectures allows a comprehensive comparison across multiple model families:

1. Linear Models
 - Logistic Regression
2. Probabilistic Models
 - Naive Bayes
3. Margin-Based Models
 - Support Vector Machine
4. Tree-Based Models

- Decision Tree
- 5. Ensemble Learning Models
 - Random Forest
 - AdaBoost
 - XGBoost / LightGBM
- 6. Deep Learning Models
 - LSTM
- 7. Transformer-Based Models
 - BERT / DistilBERT

This roadmap enables evaluation of the trade-off between interpretability, computational complexity, and predictive performance while providing a complete machine learning workflow for large-scale sentiment analysis.

Final Objective

The ultimate goal of this project is not only to identify the highest-performing model but also to understand how different machine learning paradigms interpret customer sentiment. By comparing traditional machine learning, ensemble learning, deep learning, and transformer-based approaches, this study aims to provide actionable insights for businesses seeking to leverage customer reviews for data-driven decision-making.